Retail Customer Behavior

Business analysis & portfolio report

A verified Python-to-Power BI analysis of a synthetic retail sample. The central decision is how to prioritize assortment and promotional tests without confusing descriptive patterns with business impact.

$233,081
Recorded purchase value

3,900
Unique customer records

$59.76
Average recorded purchase

PRIORITY DECISION

Test promotional incrementality before scaling spend. All subscribers received discounts, so this sample cannot isolate either effect.

THREE TAKEAWAYS

01 Concentration. Clothing contributes 44.73% of recorded revenue. Review availability and margin before changing inventory.

02 Composition. Ages 45-59 generate the most revenue and form the largest age group. Average purchase values are similar across groups.

03 Measurement. Preserve original ratings, use explicit denominators, and collect dated transactions and campaign assignment before claiming retention or uplift.

EVIDENCE STATUS

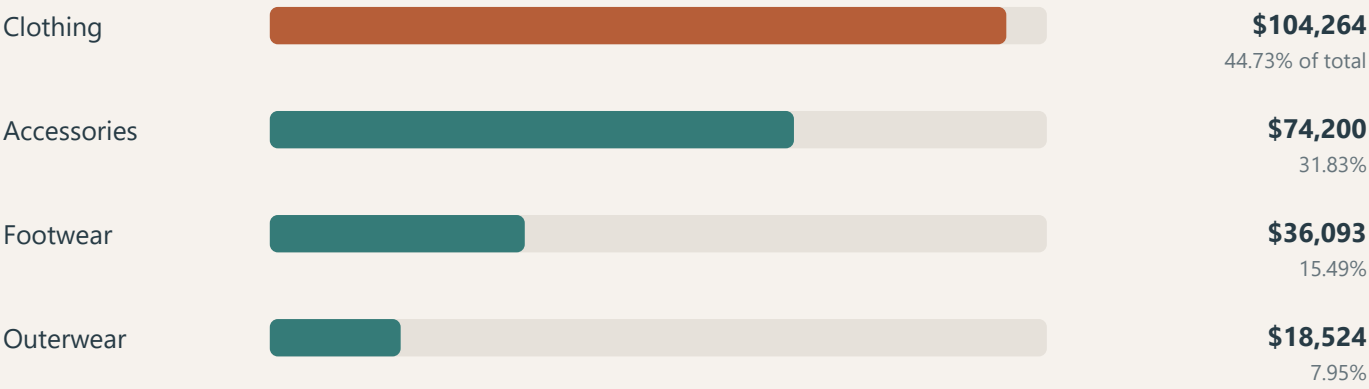
The notebook rerun, saved cleaned CSV and PBIX embedded data match across all 3,900 rows and 22 cleaned fields. Headline screenshot values reconcile at their displayed precision.

Scope: 18 raw fields, 4 engineered fields, 4 dashboard pages and 12 inspected DAX measures. Power BI Desktop refresh, rendering and live interactions were not tested in this audit.

Concentration guides investigation

Recorded revenue is concentrated in Clothing and Accessories. These are candidate areas for operational review; revenue alone does not establish profitability or unmet demand.

CATEGORY CONTRIBUTION



Source: audited cleaned dataset; total recorded purchase value = $233,081. Zero-based bars.

Top five products	Recorded revenue	Records
Blouse	$10,410	171
Shirt	$10,332	169
Dress	$10,320	166
Pants	$10,090	171
Jewelry	$10,010	171

The report uses five products here. The original dashboard displays ten; its mismatched “Top 5” title is corrected to Top 10 in the public source project.

Season	Recorded revenue	Average purchase
Fall	$60,018	$61.56
Spring	$58,679	$58.74
Winter	$58,607	$60.36
Summer	$55,777	$58.41

Decision: request category margin, returns and stockout evidence. Seasonal labels are not a dated trend and cannot support an annual sales forecast.

Separate volume from purchase value

The largest revenue segment need not spend more per purchase. The same principle applies to demographic targeting and product-rating comparisons.

Age group	Customers	Revenue	Avg. purchase
18-29	878	$52,905	$60.26
30-44	1,092	$65,216	$59.72
45-59	1,142	$68,066	$59.60
60+	788	$46,894	$59.51

Interpretation: ages 45-59 account for 29.28% of records and 29.20% of revenue. Their contribution largely reflects sample size, not unusually high purchase value.

Gender	Customers	Revenue	Avg. purchase
Female	1,248	$75,191	$60.25
Male	2,652	$157,890	$59.54

The average-purchase gap is $0.71. No significance test or demographic targeting effect was established. Compare campaign outcomes and acquisition costs before choosing a target segment.

Category	Observed ratings	Average rating / 5
Footwear	594	3.79
Accessories	1,229	3.77
Outerwear	322	3.75
Clothing	1,718	3.72

Ratings use 3,863 original observations; 37 are missing. Category medians fill a separate field only. Small rating differences and unknown missingness do not establish product-quality effects.

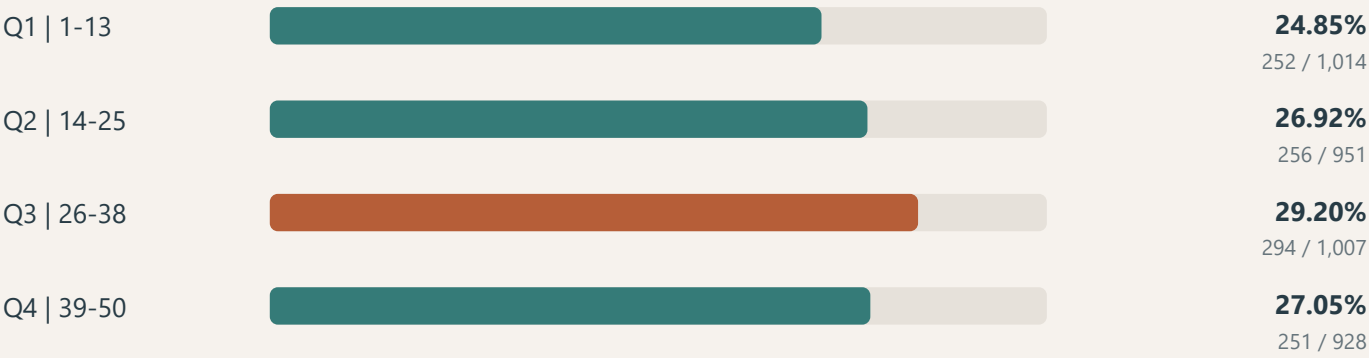
Subscriptions overlap with discounts

Subscribers represent 27% of the sample and discounted records represent 43%. Every subscriber also received a discount, leaving a key comparison cell empty.

Customer group	No discount	Discount	Total
Not subscribed	2,223	624	2,847
Subscribed	0	1,053	1,053
Total	2,223	1,677	3,900

Discounted average purchase: $59.28; undiscounted: $60.13. These groups were not randomized. The difference cannot be interpreted as a discount effect.

SUBSCRIPTION RATE BY PURCHASE-HISTORY QUARTILE



Q1-Q4 are purchase-history quartiles. Q3 leads, but rates do not increase consistently with previous-purchase counts. They do not measure retention or loyalty directly.

PROPOSED EXPERIMENT

Among eligible customers, randomly assign an offer and a holdout. Predefine the primary outcome, measurement window and minimum detectable effect before launching.

Primary outcome: incremental contribution margin per eligible customer, including discount costs and returns. Supporting outcomes: conversion, revenue per eligible customer and repeat purchases over a defined follow-up window.

Needed data: assignment, eligibility, exposure, dated orders, costs, discount amounts, returns and subscription history. No uplift target or sample size is asserted without these inputs.

Traceable evidence, explicit boundaries

PREPARATION & RECONCILIATION

Inspect 3,900 × 18 raw records → preserve observed ratings and add missingness/imputation fields → standardize names → add age bands and purchase-history quartiles → export 3,900 × 22 → reconcile with the PBIX embedded table.

Audit area	Result / boundary
Notebook and CSV	43 nonempty original code cells rerun; portable notebook executed in a fresh kernel. Export equals the original cleaned data.
PBIX inspection	One table, 22 fields and 12 DAX measures decoded. Embedded data exactly matches the notebook output.
Power BI source	Private path replaced with a parameter; Top 10 title corrected; averages use bars; rate filters preserve selections.
Verification limit	Source structure and model serialization checked. Desktop refresh, visual rendering and live DAX were not tested.

PROVENANCE & PUBLICATION

The source matches Sourav Banerjee's synthetic Customer Shopping Trends updated dataset on 17 non-rating fields and all observed ratings. The project variant has 37 ratings missing; the upstream file has none.

The publisher's metadata listed "Other (specified in description)" on 2 November 2026. Clear redistribution terms were not established, so CSVs and the data-bearing PBIX are withheld. The public portfolio provides data-free source definitions, cleared notebook outputs, documentation and original dashboard screenshots.

INTERPRETATION LIMITS

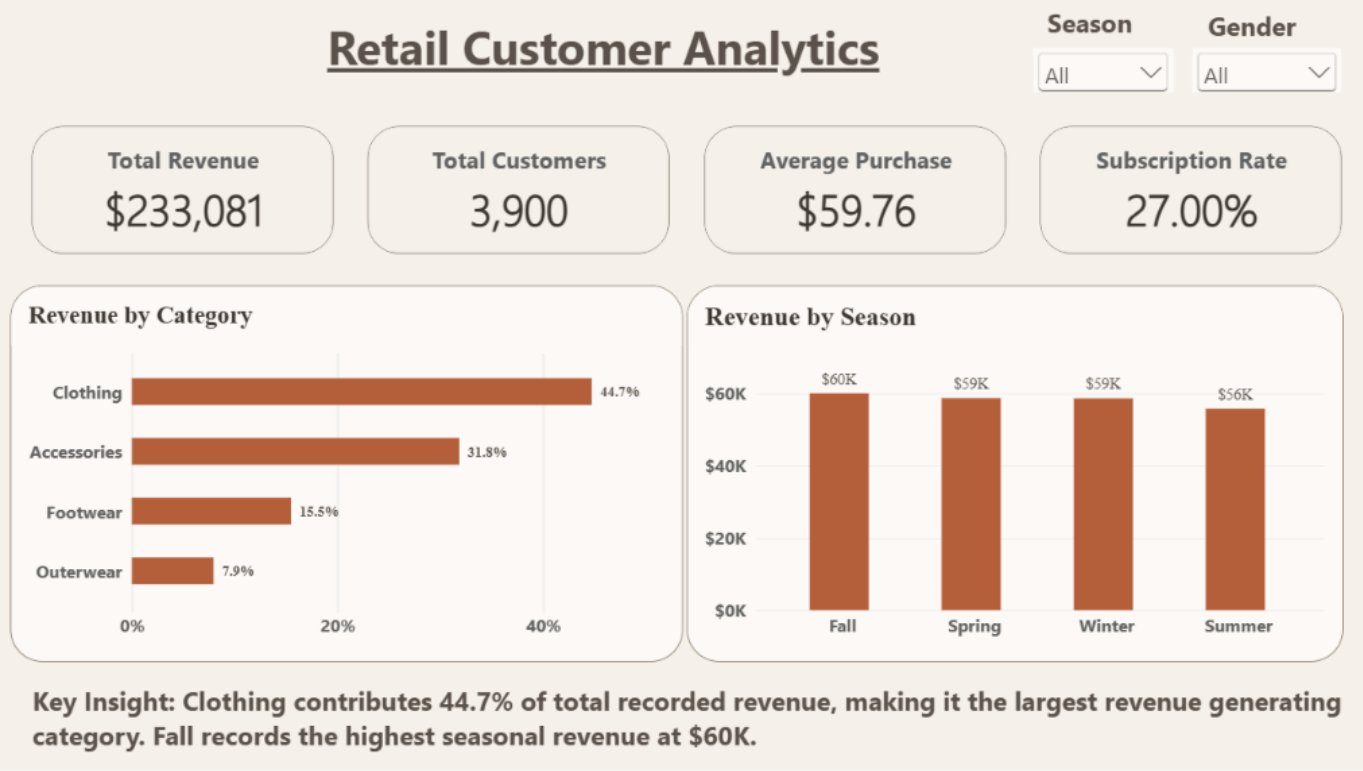
Synthetic, cross-sectional observations; no transaction dates, costs, discount amounts, campaign assignment, returns or complete purchase history. No claims of profit, growth, retention, CLV, causal uplift or population representativeness. Missing ratings may be nonrandom.

REFERENCES & DETAIL

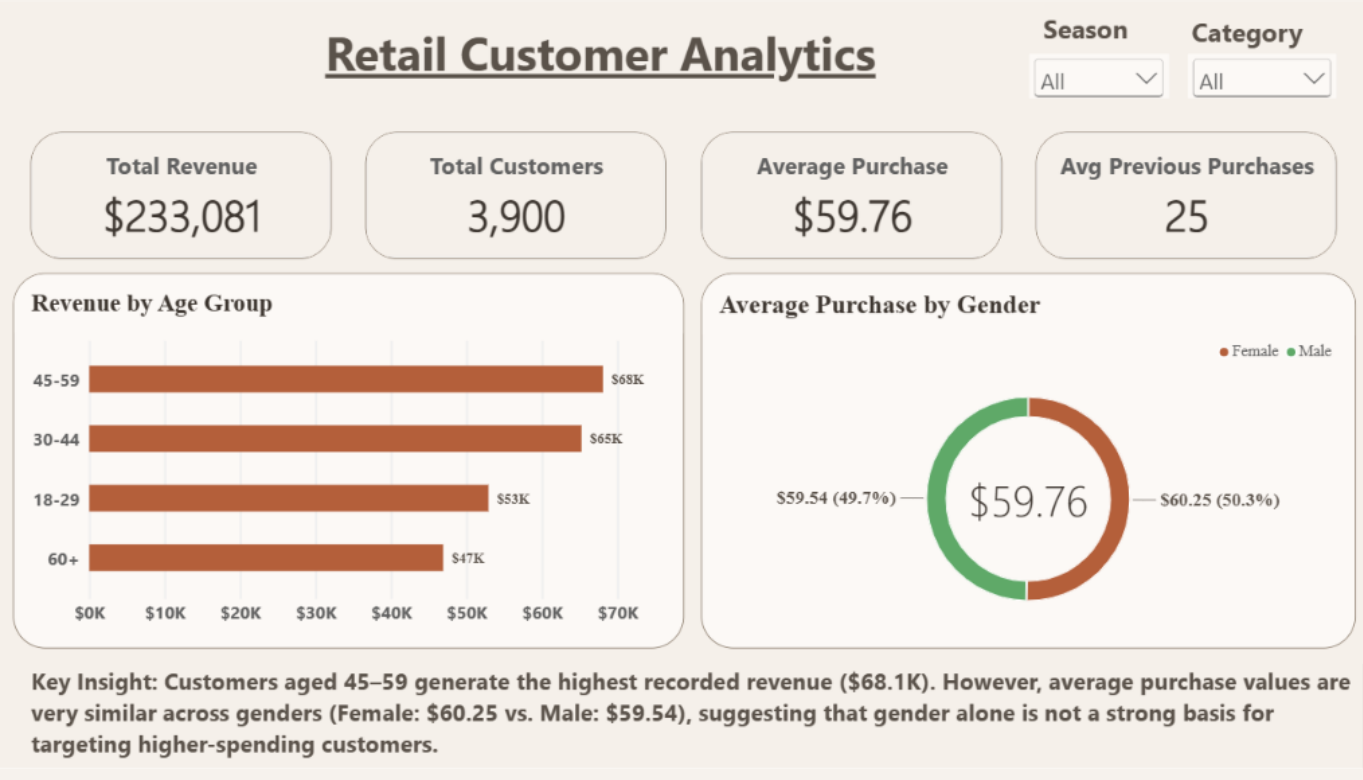
[1] Sourav Banerjee: Customer Shopping Trends Dataset
[2] Microsoft: Power BI Desktop projects · [3] Microsoft: KEEPFILTERS
Repository detail: docs/AUDIT.md, docs/DATA_DICTIONARY.md and validation/.

Dashboard evidence | overview & customers

Unaltered images from the supplied PBIX presentation. These are not Desktop-validated renders of the revised public source.



01 / Executive Overview. Original supplied snapshot; full-sample KPI values reconcile.



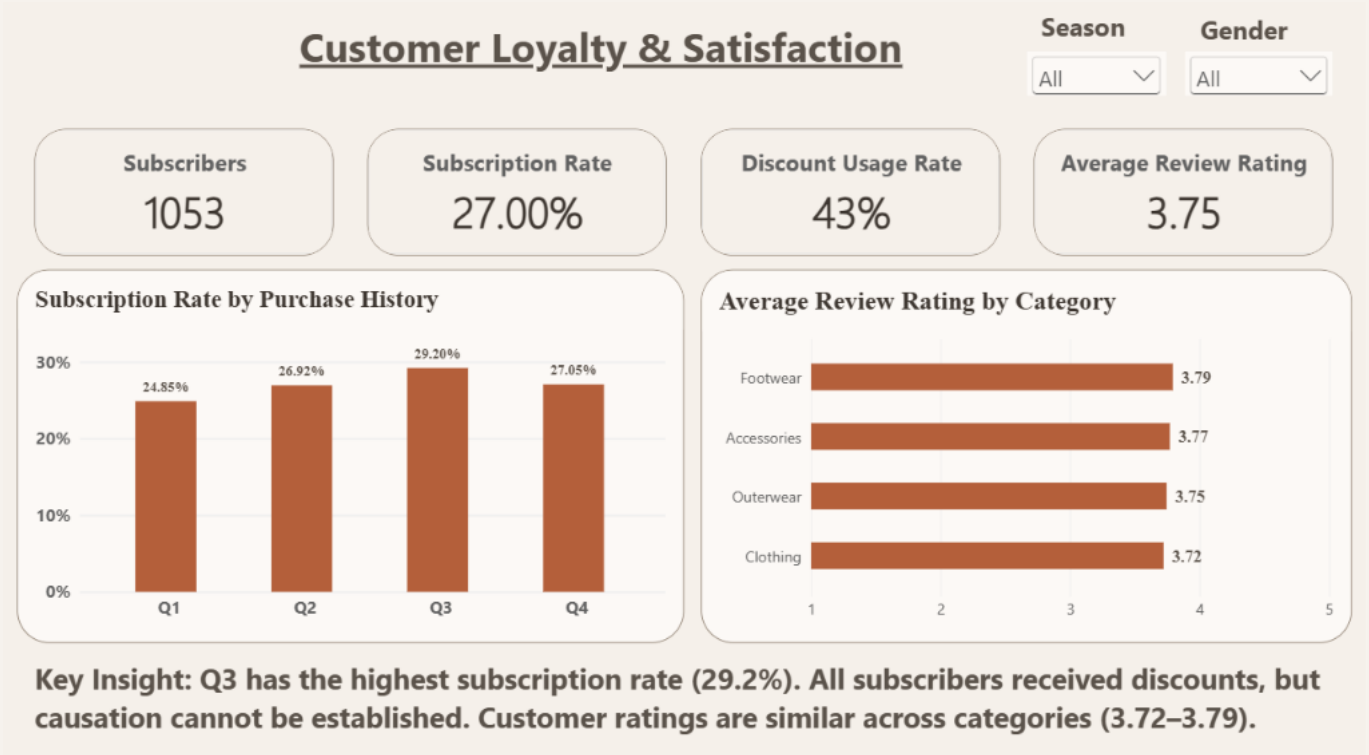
02 / Customer Insights. Original donut averages are preserved in this evidence image. The source project replaces them with a bar comparison.

Dashboard evidence | products & subscriptions

Unaltered images from the supplied PBIX presentation. These are not Desktop-validated renders of the revised public source.



03 / Product & Seasonal Performance. Original image shows ten products under a Top 5 title. The public source corrects the title to Top 10.



04 / Customer Loyalty & Satisfaction. Original ratings, subscription quartiles and full-sample promotion metrics reconcile.